

HELSINGE SODERHAMN, Sweden

WMO: 022861

Lat: 61.262N

Lon: 17.098E

Elev: 118

StdP: 14.63

Time Zone: 1.00 (EUC)

Period: 94-11

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.4	1.7	-6.1	4.0	-1.5	-2.5	4.8	2.7	22.1	32.3	19.5	31.4	6.2	300	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	79.2	64.6	75.2	63.3	71.7	61.3	66.6	75.1	64.7	71.5	63.2	69.5	9.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	89.8	68.4	62.4	84.6	67.4	60.7	79.8	66.1	31.2	74.8	29.7	71.1	28.7	69.7	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.4	17.0	15.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-8.1	68.5	4.5	2.4	-11.3	70.3	-14.0	71.7	-16.5	73.0	-19.8	74.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	24.4	22.4	29.9	39.2	45.7	55.2	61.2	59.7	51.5	42.6	33.1	26.8
	DBStd	15.23	10.16	10.61	8.00	6.09	6.36	5.45	5.73	4.92	5.28	6.40	8.12	10.22
	HDD50	4209	793	773	624	332	165	12	0	1	43	240	507	719
	HDD65	8774	1258	1193	1089	775	597	297	148	176	406	695	956	1184
	CDD50	956	0	0	0	6	33	168	348	302	88	11	0	0
	CDD65	46	0	0	0	0	0	3	31	12	0	0	0	0
	CDH74	506	0	0	0	0	0	71	328	105	2	0	0	0
	CDH80	116	0	0	0	0	0	15	85	16	0	0	0	0
Wind	WSAvg	7.0	7.8	6.8	7.1	7.0	7.1	7.0	7.0	6.5	6.7	6.8	7.2	7.4
Precipitation	PrecAvg	23.4	1.6	1.2	1.1	1.5	1.7	2.2	2.6	2.9	2.2	2.5	2.2	1.8
	PrecMax	33.6	3.3	2.4	2.5	3.8	3.5	5.1	6.2	5.6	5.3	5.9	6.3	3.5
	PrecMin	17.4	0.2	0.3	0.1	0.2	0.3	0.3	0.7	0.7	0.5	0.1	0.8	0.1
	PrecStd	3.9	0.9	0.5	0.6	1.0	0.9	1.1	1.3	1.3	1.2	1.6	1.3	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.6	44.8	55.2	68.2	71.8	82.0	88.4	82.2	73.0	61.2	53.7	48.2
		MCWB	42.5	40.2	45.7	53.5	58.1	62.8	67.3	64.2	63.0	55.0	50.8	43.6
	2%	DB	43.1	41.4	48.3	60.8	65.9	74.9	81.7	76.8	67.0	57.2	49.9	44.6
		MCWB	39.5	37.4	40.5	50.4	53.7	61.2	65.3	64.8	59.3	52.2	47.5	41.9
	5%	DB	40.9	38.8	44.5	55.5	62.4	70.2	77.2	72.0	64.1	55.0	46.6	41.5
		MCWB	38.0	35.5	38.7	47.3	52.2	59.5	63.6	63.1	57.4	50.5	45.0	39.3
	10%	DB	37.4	36.0	41.0	51.8	58.8	67.1	73.3	69.3	61.0	52.2	44.2	39.2
		MCWB	35.1	33.9	36.2	44.1	49.6	57.1	62.3	62.3	56.1	48.6	43.0	37.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.1	41.2	45.7	55.2	59.2	65.3	69.6	68.4	63.7	55.8	51.3	44.9
		MCDB	45.7	44.1	54.1	64.5	69.4	74.9	81.3	75.0	71.0	59.3	53.3	46.6
	2%	WB	40.0	38.2	41.7	51.2	55.3	62.5	67.4	66.3	60.6	53.3	47.7	42.0
		MCDB	42.4	40.5	46.7	59.6	63.1	72.6	78.0	73.1	65.5	56.5	49.1	43.5
	5%	WB	38.0	35.8	38.9	47.5	52.6	60.3	65.1	64.7	58.6	51.3	44.9	39.6
		MCDB	40.6	37.6	43.5	54.7	60.7	68.4	74.4	69.4	62.6	54.1	46.1	41.1
	10%	WB	35.0	34.0	36.7	44.6	50.5	58.2	63.3	63.5	56.8	49.5	42.9	37.6
		MCDB	37.0	35.7	40.5	50.7	57.3	65.8	71.4	68.0	60.5	51.7	43.6	39.1
Mean Daily Temperature Range	5% DB	MDBR	11.1	13.3	14.6	17.2	17.2	17.5	17.3	16.3	15.9	13.1	10.1	10.2
		MCDBR	12.0	11.9	18.1	25.8	24.0	24.2	24.1	21.0	18.9	15.2	12.0	10.4
		MCWBR	10.1	9.5	12.7	16.6	13.3	12.8	10.9	11.1	12.2	11.0	10.6	9.3
	5% WB	MCDBR	11.9	10.8	16.9	24.5	22.5	22.4	21.7	17.1	17.6	14.4	11.3	9.9
		MCWBR	10.6	9.2	12.1	16.3	13.4	13.1	11.4	10.4	12.2	11.0	10.2	9.1
Clear-Sky Solar Irradiance	taub		0.253	0.302	0.323	0.358	0.359	0.353	0.374	0.366	0.333	0.319	0.274	0.350
	taud		2.182	2.292	2.371	2.377	2.436	2.489	2.450	2.464	2.541	2.450	2.229	2.630
	Ebn at Noon		153	215	253	265	274	278	268	260	251	210	141	103
	Edh at Noon		11	20	26	31	32	31	31	29	22	17	11	8
All-Sky Solar Radiation	RadAvg		78	279	757	1227	1565	1759	1647	1260	807	366	108	46
	RadStd		11	52	87	108	165	121	167	117	102	61	24	5

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page